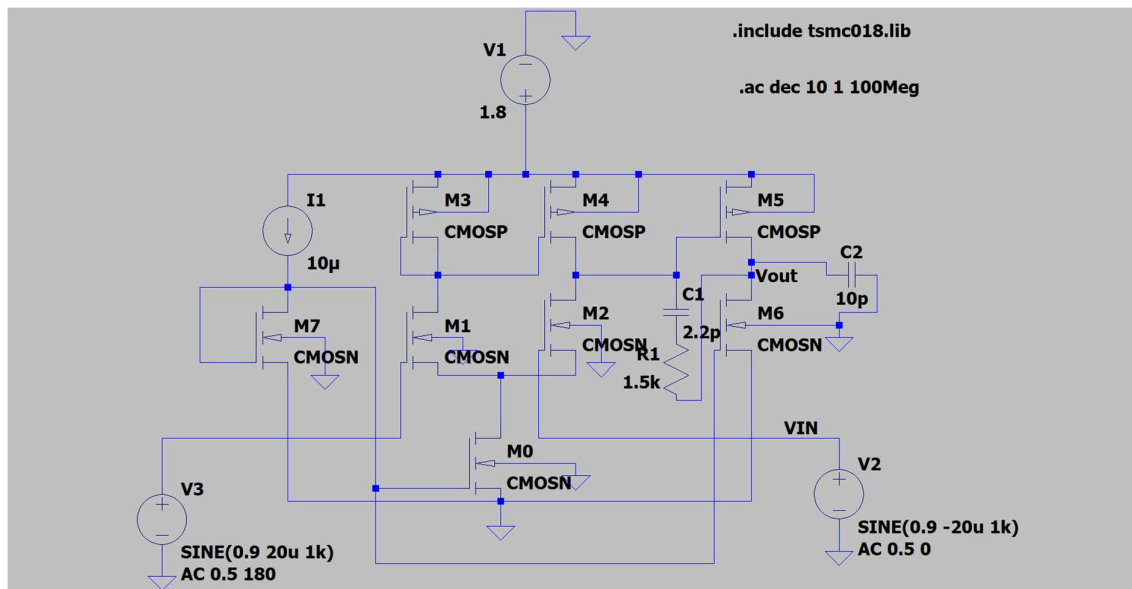
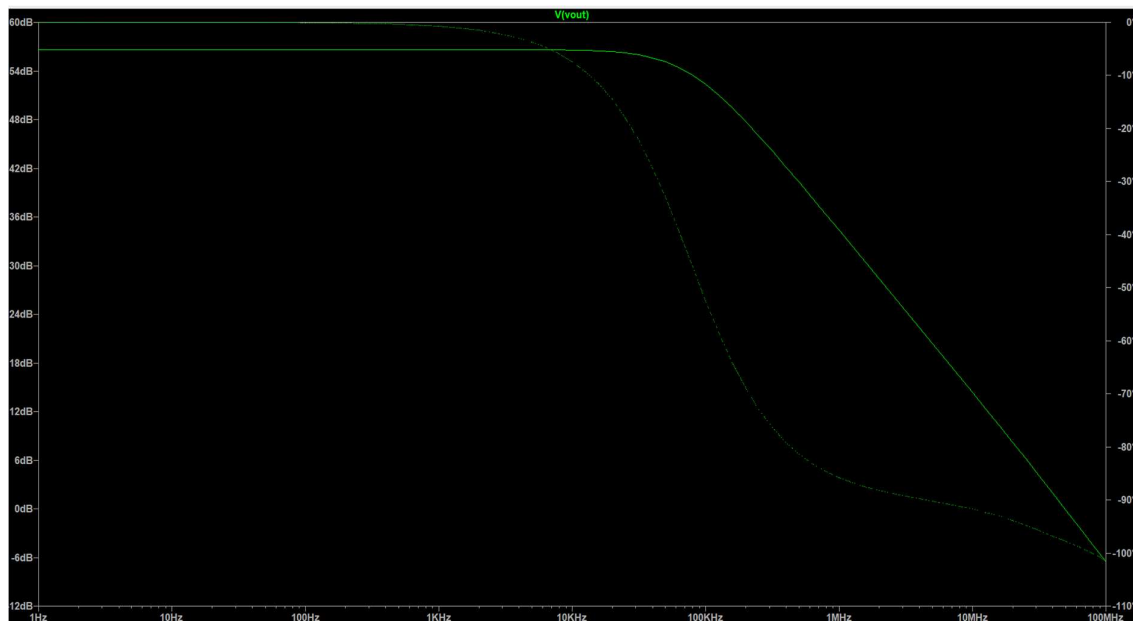


Open-Loop Design



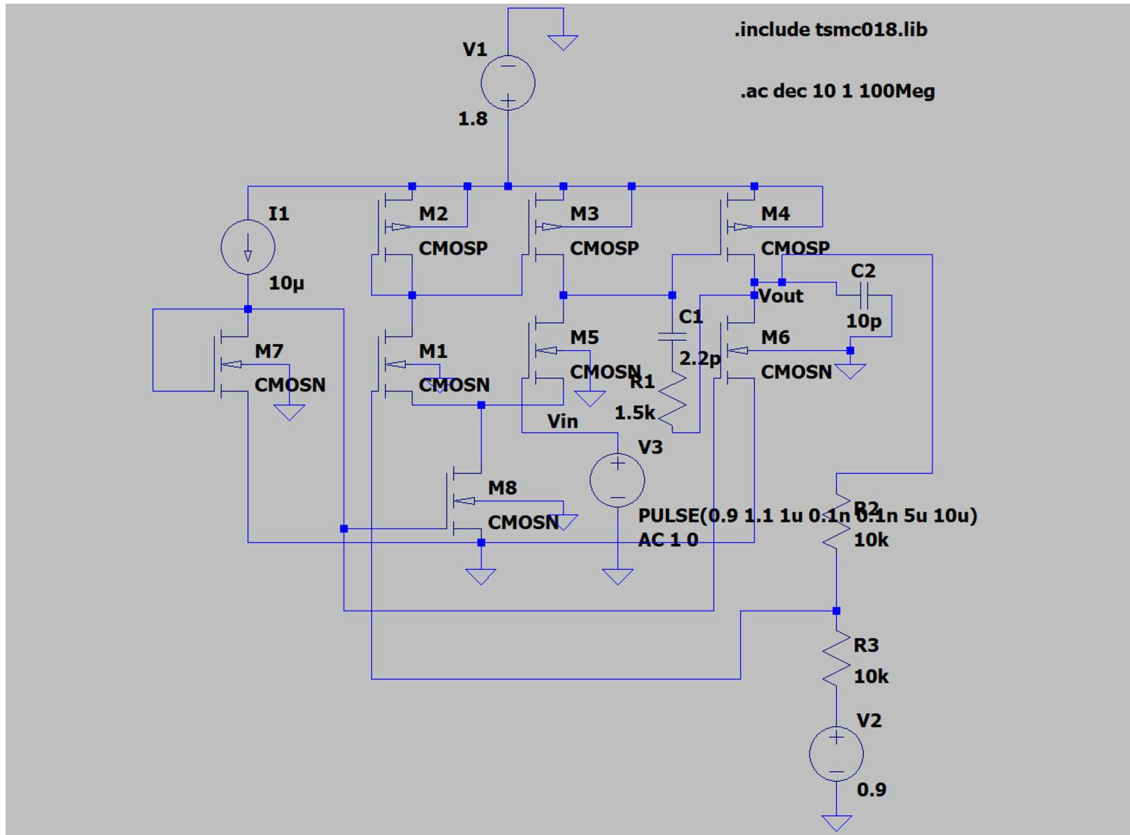
Open Loop Gain



DC GAIN: 56.627 dB

Phase Margin: $-91.667 + 180 \text{ deg} = 88.333 \text{ deg}$

Closed Loop Design

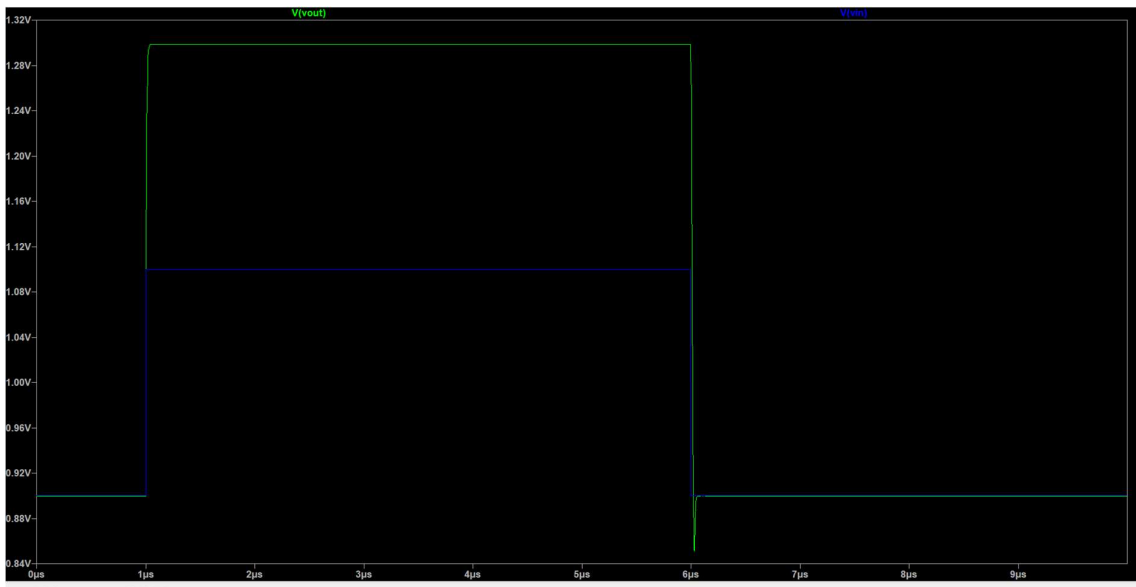


Closed Loop Gain



DC GAIN- 6 dB

Step Response:



As V_{in} rises from 0.9V to 1.1 V, V_{out} goes from 0.9 V to 1.3 V correspondingly successfully achieving a non-inverting gain of 2.

W & L tuned values:

	W	L
M0	10.86u	0.180u
M1	5.434u	0.180u
M2	5.434u	0.180u
M3	12.5u	0.180u
M4	12.5u	0.180u
M5	56u	0.180u
M6	19.566u	0.180u
M7	0.39132u	0.180u